BAMMIDI SRUTHISRI

```
from google.colab import files
import pandas as pd

uploaded = files.upload()
```

Choose files customer_p…_dataset.csv

customer_purchase_dataset.csv(text/csv) - 2796 bytes, last modified: 01/08/2026 - 100% done

```
Saving customer_purchase_dataset.csv to customer_purchase_dataset.csv
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
```

```
df = pd.read_csv("customer_purchase_dataset.csv")
```

```
df
```

	Age	AnnualIncome	WebsiteVisits	Purchased
0	25	114130	7	1
1	63	109312	10	1
2	22	33944	8	0
3	43	62122	16	1
4	24	44479	2	0
...	...	...	...	...
195	62	26557	19	0
196	29	109753	4	1
197	32	93827	7	1
198	50	94393	10	1
199	45	62944	1	0

200 rows × 4 columns

Next steps: Generate code with df | New interactive sheet

```
df.head()
```

	Age	AnnualIncome	WebsiteVisits	Purchased
0	25	114130	7	1
1	63	109312	10	1
2	22	33944	8	0
3	43	62122	16	1
4	24	44479	2	0

Next steps: Generate code with df | New interactive sheet

```
df.tail()
```

	Age	AnnualIncome	WebsiteVisits	Purchased
195	62	26557	19	0
196	29	109753	4	1
197	32	93827	7	1
198	50	94393	10	1
199	45	62944	1	0

```
df.shape
```

```
(200, 4)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Age            200 non-null    int64
 1   AnnualIncome   200 non-null    int64
 2   WebsiteVisits  200 non-null    int64
 3   Purchased      200 non-null    int64
dtypes: int64(4)
memory usage: 6.4 KB
```

```
df.isnull().sum()
```

	0
Age	0
AnnualIncome	0
WebsiteVisits	0
Purchased	0

dtype: int64

```
df.describe()
```

	Age	AnnualIncome	WebsiteVisits	Purchased
count	200.000000	200.00000	200.000000	200.000000
mean	40.910000	71968.28500	10.735000	0.660000
std	13.764783	29470.71412	5.905859	0.474898
min	18.000000	20685.00000	1.000000	0.000000
25%	30.000000	46016.75000	5.000000	0.000000
50%	39.500000	72220.50000	11.000000	1.000000
75%	51.000000	96874.50000	16.000000	1.000000
max	65.000000	119595.00000	20.000000	1.000000

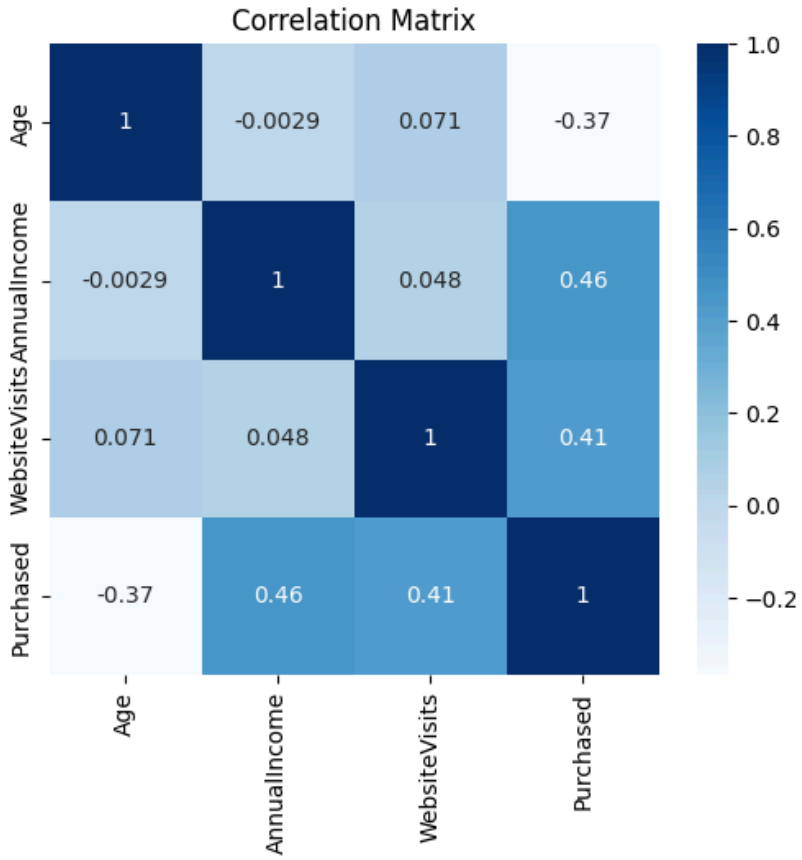
```
df.duplicated().sum()
```

```
np.int64(0)
```

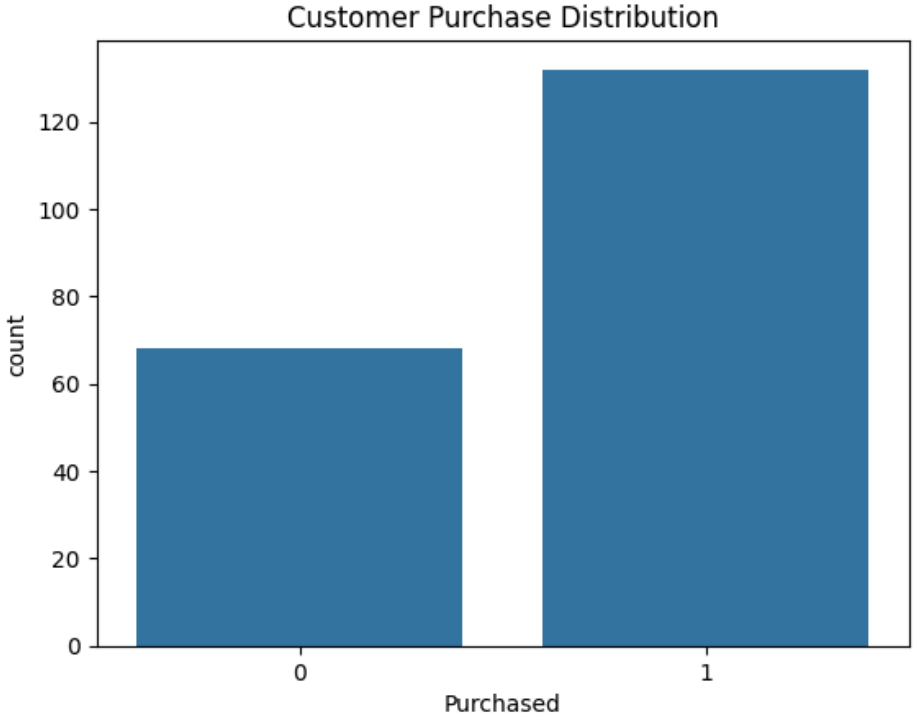
```
df.corr()
```

	Age	AnnualIncome	WebsiteVisits	Purchased
Age	1.000000	-0.002943	0.070916	-0.367548
AnnualIncome	-0.002943	1.000000	0.048473	0.462331
WebsiteVisits	0.070916	0.048473	1.000000	0.412054
Purchased	-0.367548	0.462331	0.412054	1.000000

```
plt.figure(figsize=(6,5))
sns.heatmap(df.corr(), annot=True, cmap='Blues')
plt.title("Correlation Matrix")
plt.show()
```




```
sns.countplot(x='Purchased', data=df)
plt.title("Customer Purchase Distribution")
plt.show()
```



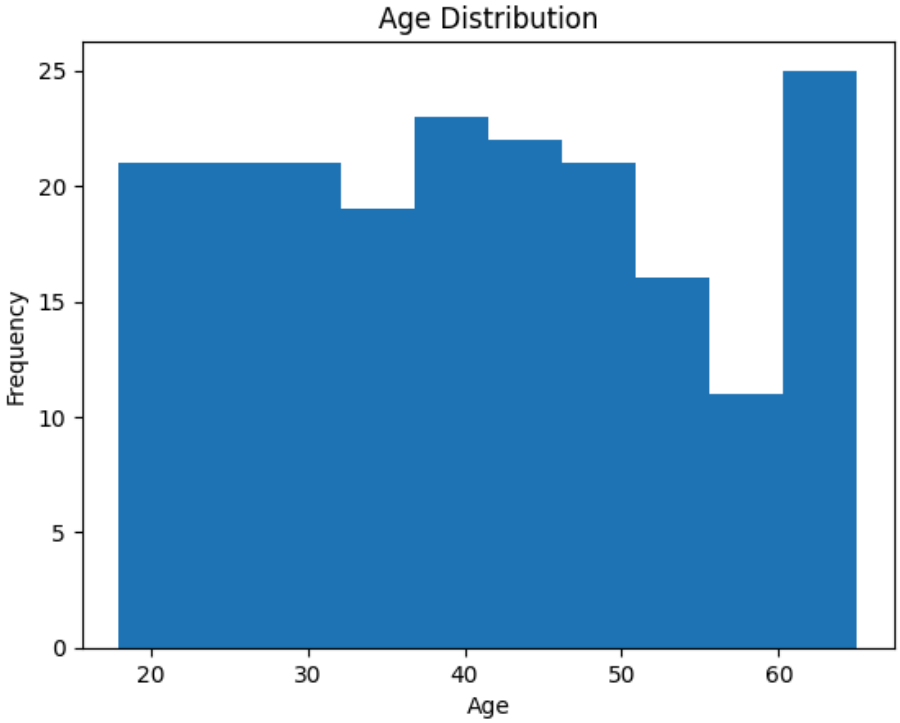

```
plt.hist(df['Age'], bins=10)

plt.title("Age Distribution")

plt.xlabel("Age")

plt.ylabel("Frequency")

plt.show()
```



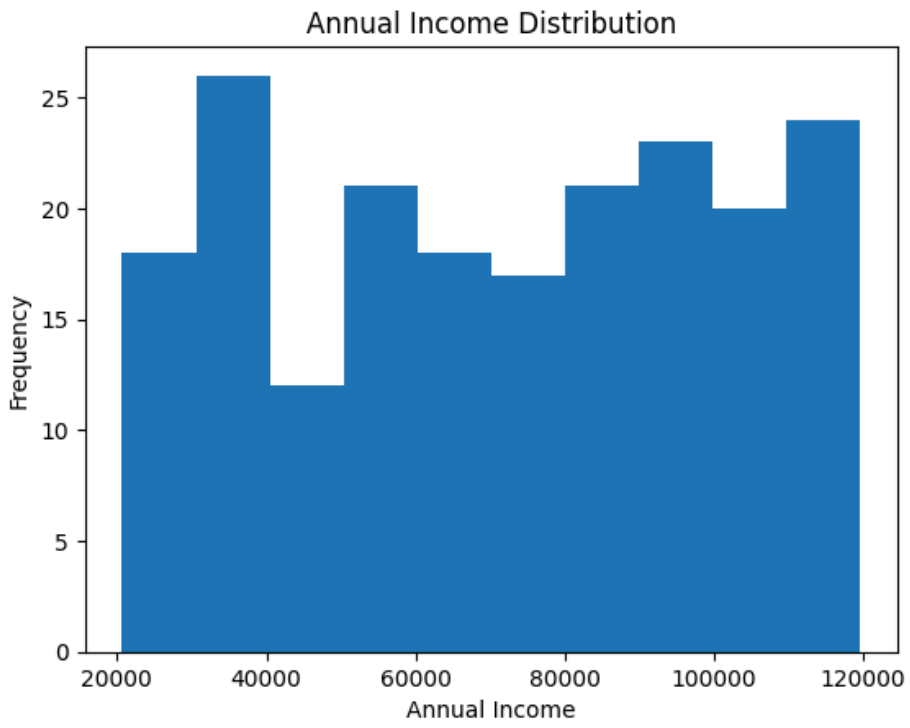

```
plt.hist(df['AnnualIncome'], bins=10)

plt.title("Annual Income Distribution")

plt.xlabel("Annual Income")
```

```
plt.ylabel("Frequency")

plt.show()
```



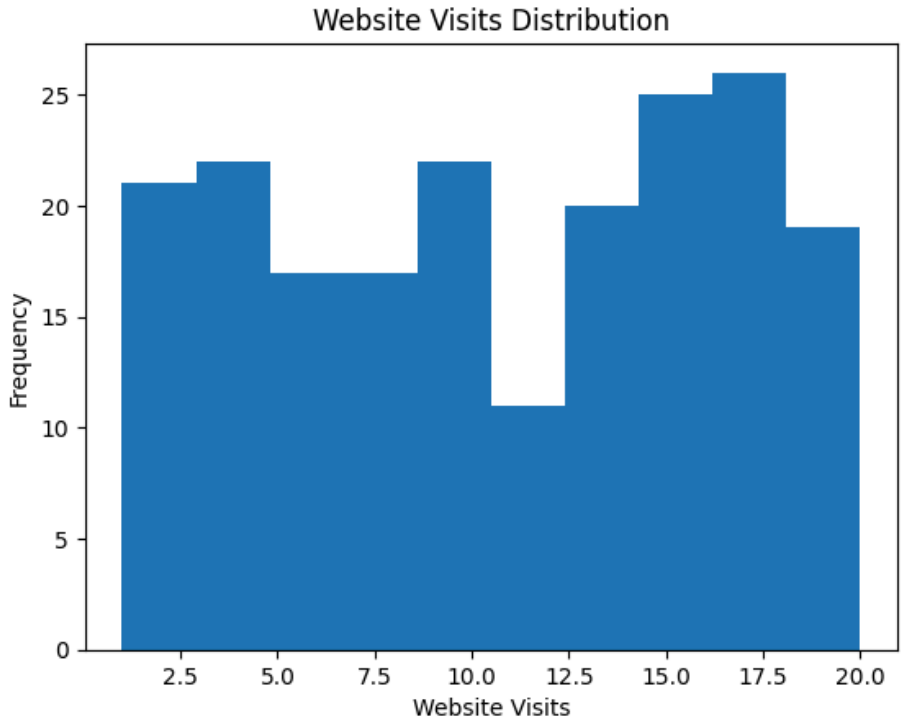

```
plt.hist(df['WebsiteVisits'], bins=10)

plt.title("Website Visits Distribution")

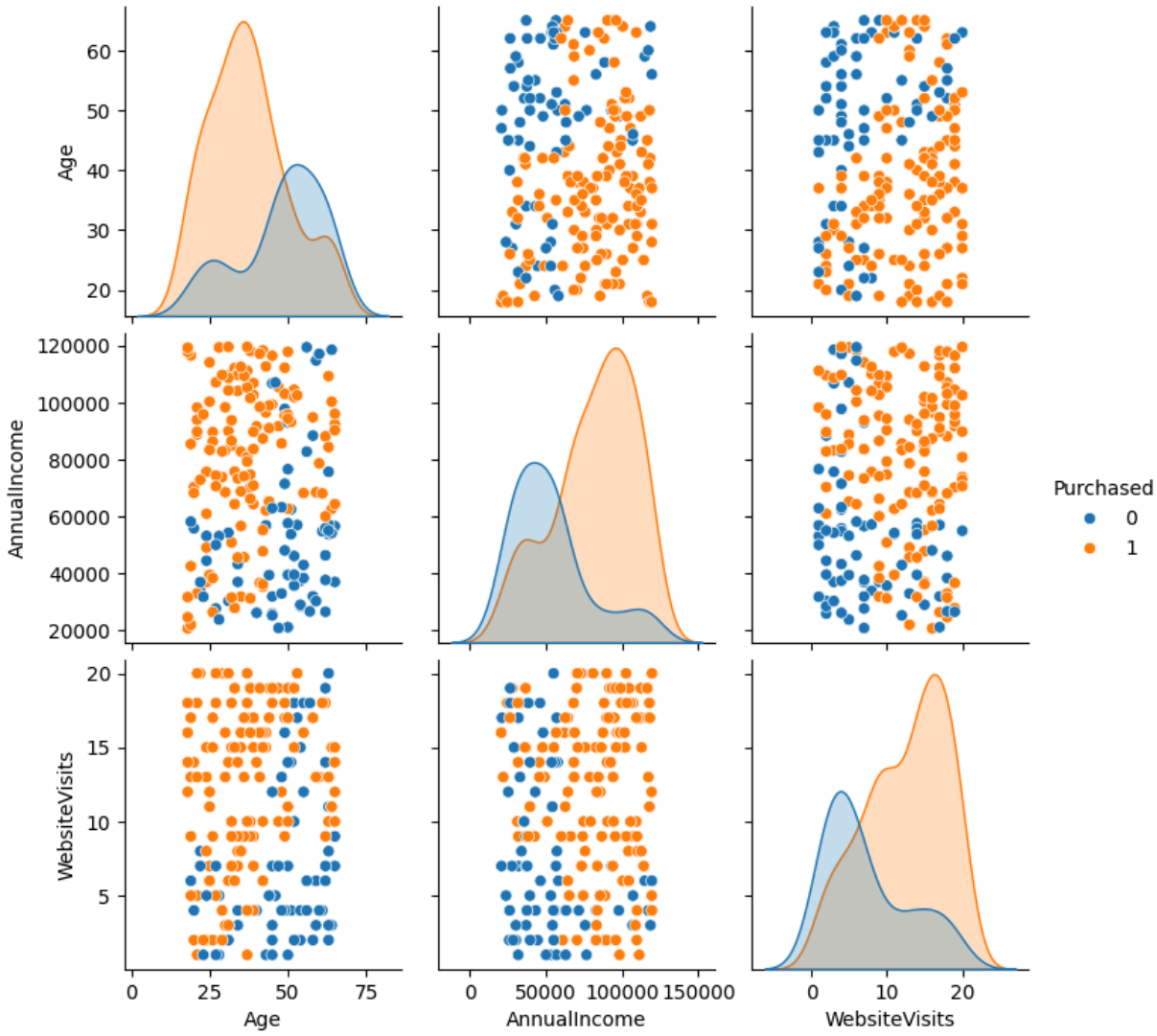
plt.xlabel("Website Visits")

plt.ylabel("Frequency")

plt.show()
```



```
sns.pairplot(df, hue='Purchased')
plt.show()
```




```
X = df[['Age','AnnualIncome','WebsiteVisits']]
```

```
y = df['Purchased']
```

```
X.head()
```

	Age	AnnualIncome	WebsiteVisits
0	25	114130	7
1	63	109312	10
2	22	33944	8
3	43	62122	16
4	24	44479	2

Next steps: Generate code with X | New interactive sheet

```
y.head()
```

	Purchased
0	1
1	1
2	0
3	1
4	0

dtype: int64

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(160, 3)
(40, 3)
(160,)
(40,)
```

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy =", accuracy)
```

```
Accuracy = 0.875
```

```
cm = confusion_matrix(y_test, y_pred)

print(cm)
```

```
[[ 7  4]
 [ 1 28]]
```

```
plt.figure(figsize=(5,4))

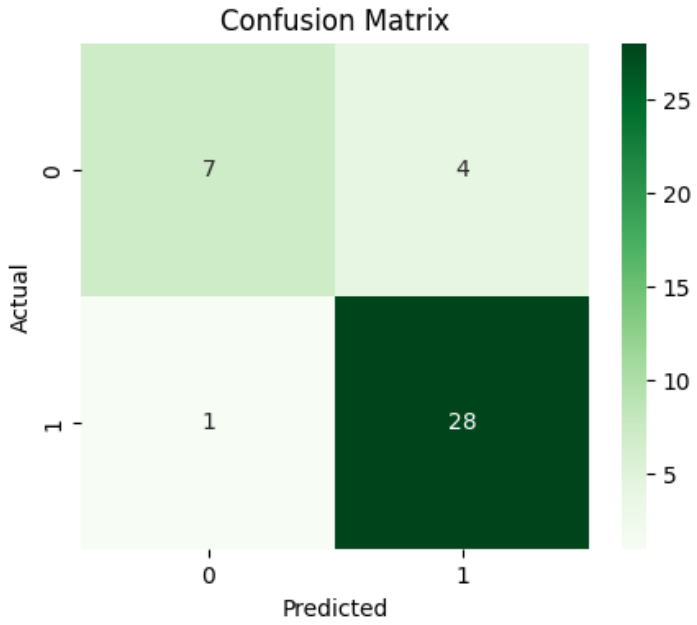
sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Greens')

plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")

plt.title("Confusion Matrix")

plt.show()
```




```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.88      0.64      0.74        11
           1       0.88      0.97      0.92        29

    accuracy                           0.88        40
   macro avg       0.88      0.80      0.83        40
weighted avg       0.88      0.88      0.87        40
```

```
print("Coefficients:")

print(model.coef_)
```

```
Coefficients:
[[-1.38643716e-01  7.99259375e-05  4.03517512e-01]]
```

```
print("Intercept:")

print(model.intercept_)
```

```
Intercept:
[-2.59644069]
```

```
customer = pd.DataFrame({
    'Age': [30],
    'AnnualIncome': [70000],
    'WebsiteVisits': [15]
})

prediction = model.predict(customer)

print("Prediction:", prediction)
```

```
Prediction: [1]
```

```
if prediction[0] == 1:
    print("Customer is likely to purchase the product.")
else:
    print("Customer is not likely to purchase the product.")
```

```
Customer is likely to purchase the product.
```

```
probability = model.predict_proba(customer)

print("Prediction Probability:")

print(probability)
```

```
Prediction Probability:
[[0.00745175 0.99254825]]
```

```
results = pd.DataFrame({
    'Actual': y_test,
    'Predicted': y_pred
})

results.head()
```

	Actual	Predicted
95	0	0
15	1	1
30	1	1
158	1	1
128	1	1

Next steps: Generate code with `results` New interactive sheet

```
results.to_csv("Customer_Purchase_Predictions.csv", index=False)

print("Predictions saved successfully.")
```

```
Predictions saved successfully.
```